

Chapter 13: Gestational Trophoblastic Disease —

Revision Notes

Williams Obstetrics 26e, pp. 614–639 of the PDF. High-yield numbers are in **bold**. The five tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: GTD = tumors of abnormal trophoblast proliferation, all of which make **hCG**. Serum hCG is therefore the key to **diagnosis, treatment and surveillance**. Molar pregnancies are **benign but can progress to GTN**. GTN is usually diagnosed from **hCG behavior rather than histology** and is **highly curable with chemotherapy**.

1. Classification

- **Hydatidiform moles** have **villi**: excessively edematous immature placentas.
 - **Benign**: complete and partial mole.
 - **Malignant: invasive mole** (penetrates and destroys the myometrium; can metastasize).
- **Nonmolar trophoblastic neoplasms** have **no villi**: choriocarcinoma, placental site trophoblastic tumor (PSTT), epithelioid trophoblastic tumor (ETT). They are distinguished by the trophoblast type they contain.
- **Gestational trophoblastic neoplasia (GTN)** = the malignant forms: **invasive mole, choriocarcinoma, PSTT, ETT**.
 - Can arise weeks or years after **any** pregnancy, but most often after a mole.
 - Rarely diagnosed histologically → managed as **one clinical entity** based on hCG and clinical findings.

Table 13-1. Modified WHO Classification of GTD

Molar Pregnancies	Trophoblastic Tumors
Hydatidiform mole: Complete	Choriocarcinoma
Hydatidiform mole: Partial	Placental site trophoblastic tumor
Invasive mole	Epithelioid trophoblastic tumor

GTN = gestational trophoblastic disease; WHO = World Health Organization. From Hui, 2014a,b.

2. Hydatidiform Mole (Molar Pregnancy)

- Classic histology: **trophoblast proliferation + villous stromal edema**.
- Classified as complete or partial by the degree of histological change, **karyotype, immunostaining**, and presence of **embryonic elements**. **GTN follows complete moles more often**.
- **Gross**: complete mole = mass of **clear vesicles** of varying size hanging in clusters from thin pedicles. Partial mole = **focal**, less advanced change with some **fetal tissue**. Both usually fill the cavity; rarely ectopic.

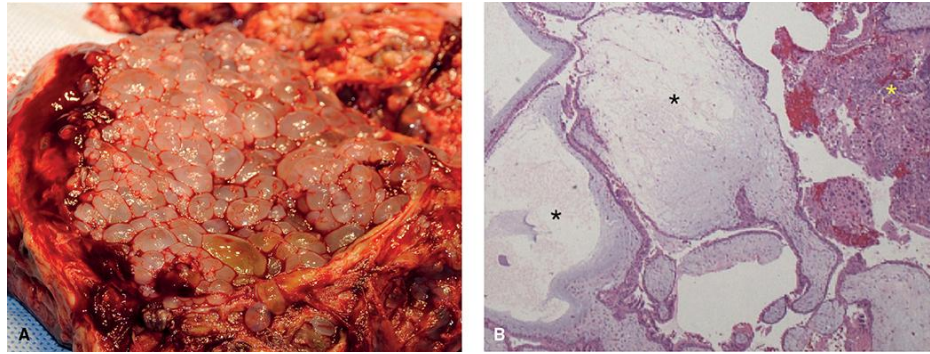


Figure 13-1. Complete mole: (A) grape-like vesicles of variable size; (B) edematous avascular villi with central cisterns (black asterisks) and haphazard trophoblastic hyperplasia (yellow asterisk).

Table 13-2. Features of Partial and Complete Hydatidiform Moles

Feature	Partial Mole	Complete Mole
Karyotype^a	69,XXX or 69,XXY	46,XX
Clinical presentation		
Diagnosis	Missed abortion	Molar gestation
Uterine size	Small for dates	Large for dates
Theca-lutein cysts	Rare	25–30% of cases
Initial hCG levels	<100,000 mIU/mL	>100,000 mIU/mL
Medical complications ^b	Rare	Uncommon
Rate of subsequent GTN	1–5% of cases	15–20% of cases
Pathology		
Embryo-fetus	Often present	Absent
Amnion, fetal erythrocytes	Often present	Absent
Villous edema	Focal	Widespread
Trophoblastic proliferation	Focal, slight to moderate	Slight to severe
Trophoblast atypia	Mild	Marked
p57 ^{KIP2} immunostaining	Positive	Negative

^aTypical karyotypes. ^bThese include anemia, hyperthyroidism, hyperemesis gravidarum, preeclampsia, and infection. GTN = gestational trophoblastic neoplasia; hCG = human chorionic gonadotropin.

Epidemiology and risk factors

- Higher in **Asians, Hispanics and American Indians**.
- US/Europe incidence **~1 per 1000 deliveries** (stable).
- **Strongest risk factors: maternal age (both extremes) and a prior mole.**

Prior history	Risk of another mole
One complete mole	0.9%
One partial mole	0.3%
Two complete moles	~20%

Pathogenesis

	Complete mole	Partial mole
Ploidy	Diploid	Triploid
Usual karyotype	46,XX	69,XXX or 69,XXY (69,XYY rare)
Mechanism	Androgenesis : empty/inactivated ovum + one haploid sperm that duplicates its chromosomes. Less often dispermy (46,XY or 46,XX)	Dispermy (two sperm + one normal haploid ovum = diandry); less often one unreduced diploid 46,XY sperm
Parental origin	All paternal	Two paternal sets + one maternal
Fetus	None	Some embryonic development, but lethal : severe FGR, multiple anomalies

Memory hook: "Complete = Completely paternal (46,XX, no fetus). Partial = Part mother (69, triploid, with fetus)."

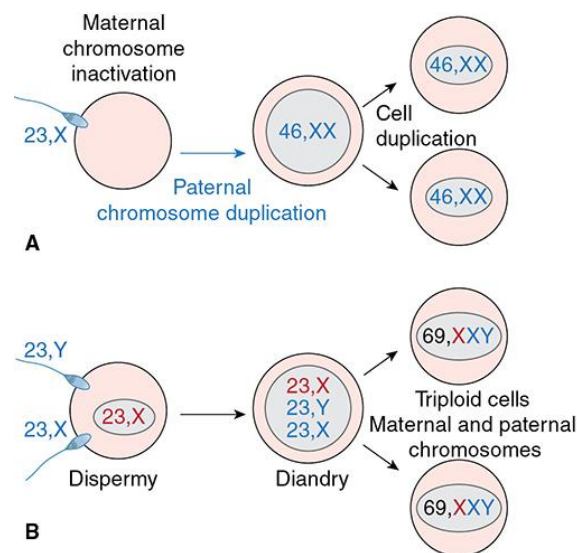


Figure 13-2. (A) Complete mole: a 23,X sperm fertilizes an inactivated egg and duplicates → 46,XX, all paternal. (B) Partial mole: dispermy of a normal egg → 69,XXY triploid (diandry).

Twin pregnancy: normal fetus + complete mole

- Must be distinguished from a **single partial mole** with an abnormal fetus. Other mimics: **placental mesenchymal dysplasia**, subchorionic hematoma, chorioangioma.
- Confirm with **CVS, amniocentesis or fetal blood sampling + fetal karyotype**.
- Complications: **thyrotoxicosis** (common); most worrying are **preeclampsia and hemorrhage** → often **preterm delivery**. Many choose termination.
- Of 92 continued pregnancies: **42%** miscarried or had a perinatal death; **~60%** delivered preterm; **40%** at term.
- **GTN risk is not significantly different** whether the pregnancy is continued or terminated.

Clinical findings

- Diagnosed earlier today (early prenatal care, universal sonography): median evacuation at **9 wk (complete)** and **12 wk (partial)** → most treated before complications.
- Usually **1–2 months of amenorrhea** first; symptoms are more marked with complete moles.
- **Uterine bleeding** in almost all untreated moles (spotting to profuse; often intermittent for weeks); concealed bleeding → **iron-deficiency anemia**.
- **Nausea and vomiting** may be significant.

- **Uterus larger than dates and softer.** No fetal heart motion with a complete mole.
- **Theca-lutein cysts** (multiple, ovaries full and cystic): more common with complete moles; from **hCG overstimulation** of the theca layer (hCG and LH share a receptor).
 - They **regress after evacuation** → **expectant management**.
 - Torsion/infarction/hemorrhage can occur; **oophorectomy only if extensive infarction persists after untwisting**.
- **Thyroid:** hCG is thyrotropin-like → **↑ free T4, ↓ TSH**; clinical thyrotoxicosis is unusual (bleeding and sepsis may mimic it); fT4 normalizes rapidly after evacuation. "Thyroid storm" reported.
- **Severe preeclampsia/eclampsia** is relatively common with **advanced** moles (rare now), except with a coexisting normal twin.

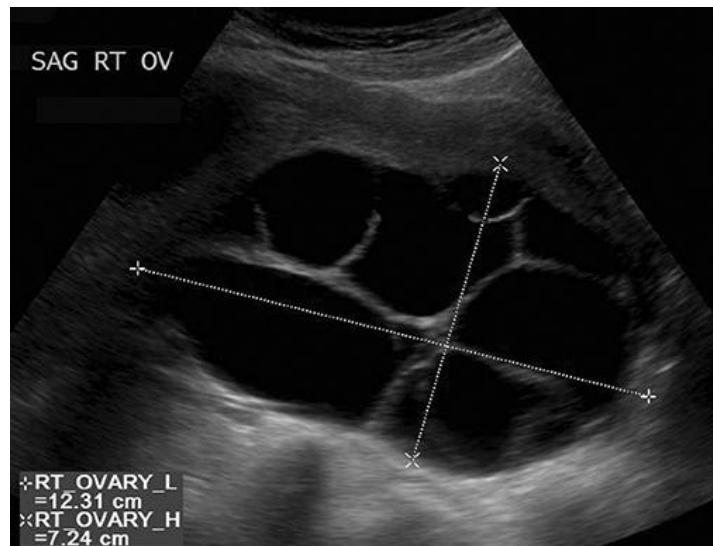


Figure 13-3. Theca-lutein cysts: an enlarged (12 × 7 cm) multiloculated ovary from hCG overstimulation with a complete mole; managed expectantly, they regress after evacuation.

3. Diagnosis

Serum β -hCG

- **Complete mole:** β -hCG usually **above** expected for gestational age; **values in the millions** with advanced moles.
- **⚠ Hook effect:** very high hCG saturates the assay antibody → **false-negative urine test**. Serum hCG, with or without **dilution**, is positive.
- **Partial mole:** may be high, but more often **within the expected range**.

Sonography

- The mainstay, but in >1000 moles **sensitivity 44%, specificity 74%**.
- **Complete mole:** echogenic mass filling the cavity with numerous anechoic cysts, **no fetus or amnionic sac** = "snowstorm". Color Doppler: marked **surrounding myometrial** flow but **no internal flow** (avascular villi).
- **Partial mole:** **thick, multicystic placenta + fetus or fetal tissue**; thin septa in the sac. Fetus usually dies in the 1st trimester; survivors show **FGR, oligohydramnios, limb or CNS defects**.
- **Early moles:** <1/2 show these features. An early complete mole may be a **polypoid hyperechoic mass without cysts**, surrounded by fluid.
- **Mimics:** incomplete or missed abortion (most common); multifetal pregnancy; leiomyoma with cystic degeneration.

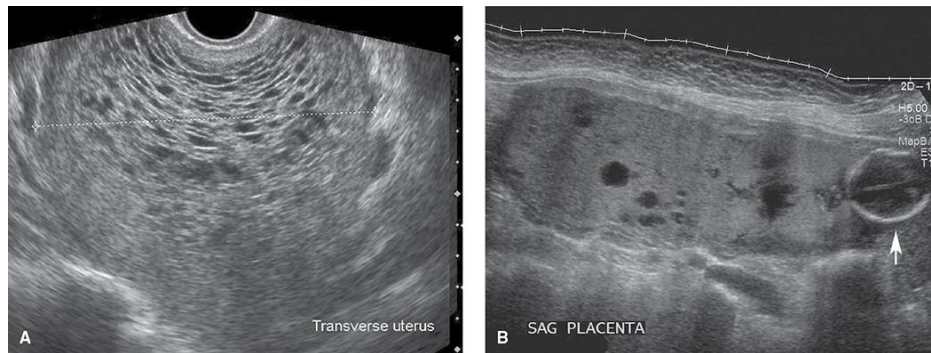


Figure 13-4. (A) Complete mole: "snowstorm" echogenic mass with many small cysts and no fetus. (B) Partial mole: enlarged multicystic placenta next to a fetal head (arrow).

Pathology

- Moles must be distinguished from **hydropic abortuses** (normal biparental conception that failed; villi swollen). **These need no surveillance.**
- **Before 10 weeks** classic molar changes may be absent (villi not enlarged, stroma not yet edematous).
- **p57^{KIP2} immunostaining:**
 - p57 gene is **paternally imprinted, maternally expressed** → protein present only when a **maternal allele** exists.
 - **Complete mole: p57 NEGATIVE** (no maternal genome).
 - **Positive** in normal placenta, hydropic abortus and **partial mole**.
- **Molecular genotyping** (parental source of alleles) separates a partial mole from a hydropic abortus:

Genotype	Diagnosis
Diploid diandric	Complete mole
Triploid diandric-monogynic	Partial mole
Biparental diploid	Nonmolar hydropic abortus

4. Management

- Maternal death is rare thanks to early diagnosis, evacuation and surveillance.
- **Preoperative evaluation** looks for preeclampsia, hyperthyroidism, anemia, electrolyte loss from hyperemesis and metastases.
- **Chest radiograph** for most. **CT/MRI only** if the CXR shows lung lesions or other extrauterine disease is suspected.

Table 13-3. Some Considerations for Management of Hydatidiform Mole

Preoperative

- Laboratory:
 - Hemogram; serum β -hCG, creatinine, and hepatic aminotransferase levels
 - **TSH, free T₄** levels
 - Type and Rh; group & screen or crossmatch depending on uterine size
- **Chest radiograph**
- Consider hygroscopic dilators

Intraoperative

- Large-bore intravenous catheter(s)
- Regional or general anesthesia

- **Oxytocin (Pitocin): 20 units in 1000 mL Ringer lactate** for continuous infusion
- One or more other uterotonic agents may be added as needed:
 - **Methylergonovine** (Methergine): **0.2 mg** = 1 mL = 1 ampule IM every 2 h prn
 - **Carboprost tromethamine** (PGF_{2α}) (Hemabate): **250 µg** = 1 mL = 1 ampule IM every 15–90 min prn
 - **Misoprostol** (PGE₁) (Cytotec): **200-µg** tablets for rectal administration, **800–1000 µg** once
- **Karman cannula: size 10 or 14 mm**
- Consider sonography machine

Postevacuation

- **Anti-D immune globulin** (Rhogam) if Rh D-negative
- Initiate effective contraception^a
- Review pathology report
- **Serum hCG levels: within 48 h of evacuation, weekly until undetectable, then monthly for 6 months**

^aIntrauterine devices are not suitable during surveillance. hCG = human chorionic gonadotropin; IM = intramuscular; PG = prostaglandin; T₄ = thyroxine; TSH = thyroid-stimulating hormone. The printed table gives the misoprostol dose as "200 mg tablets ... 800–1000 mg once", an obvious typo for µg (corrected here).

Molar evacuation

- **Suction curettage is preferred regardless of uterine size.**
- **Osmotic dilator** first if the cervix is minimally dilated.
- Bleeding can exceed that of a nonmolar uterus of the same size → with large moles ensure **adequate anesthesia, IV access and blood bank support**.
- Dilate to admit a **large Karman cannula (10–14 mm)**. **Start oxytocin as evacuation begins**. Intraoperative **sonography** helps ensure an empty cavity and ↓ perforation.
- After the myometrium contracts, gentle **sharp curettage with a large-loop Sims curette**.
- Ongoing bleeding → more uterotonics → rarely **pelvic artery embolization, packing, hysterectomy**.
- **Trophoblastic deportation** into pelvic veins during evacuation → with large moles, **respiratory insufficiency, pulmonary edema or embolism**; usually clears rapidly without specific treatment.
- **Anti-D immunoglobulin for D-negative women** (partial moles may contain fetal D-positive red cells; complete vs partial is not known until histology).
- **Prophylactic chemotherapy is NOT recommended** (no long-term benefit, significant toxicity).
- **Hysterectomy with ovarian preservation:** for women who have finished childbearing with **high-risk complete moles**. **Women aged 40–49: up to 50% develop GTN**, and hysterectomy markedly ↓ this. Theca-lutein cysts are left (they regress).
- **Labor induction or hysterotomy** is rarely used: more blood loss and possibly more persistent disease.

Postevacuation surveillance

- Use an hCG assay that detects **all hCG isoforms** (different from routine pregnancy tests).
- **Baseline β-hCG within 48 h** → repeat every **1–2 weeks** until **undetectable**.
 - Median time to resolution: **6 wk (partial), 7 wk (complete)**.
 - Normal by **14 wk in 95%** and by **25 wk in 99%**.
- Then **monthly for 6 more months** → stop surveillance and allow pregnancy.
- **Reliable contraception** throughout: **combined hormonal contraception, DMPA or progestin implant** (last two if poor compliance expected).
 - ⚠ **No IUD until hCG is undetectable** (perforation risk if an invasive mole is present).
- Pregnancy during surveillance is not advised, but live-birth and anomaly rates look normal.
- **Rising or plateaued hCG (not pregnant) → evaluate for GTN.**

Risk factors for postmolar GTN

- **Complete mole (15–20%)** vs partial mole (1–5%). Earlier evacuation has **not** lowered this risk.
- **Older maternal age**
- **Pre-evacuation β -hCG >100,000 mIU/mL**
- **Uterus large for gestational age**
- **Theca-lutein cysts >6 cm**
- **Slow β -hCG decline**

5. Gestational Trophoblastic Neoplasia

- Invasive mole, choriocarcinoma, PSTT, ETT: almost always follow a recognized pregnancy.

Antecedent pregnancy	Share of GTN
Hydatidiform mole	1/2
Miscarriage or tubal pregnancy	1/4
Preterm or term pregnancy	1/4

- Usually diagnosed from **persistently raised β -hCG alone** (tissue rarely available).

Table 13-4. Criteria for Diagnosis of Gestational Trophoblastic Neoplasia

1. **Plateau** of β -hCG level ($\pm$ **10 percent**) for **four measurements** during a period of **3 weeks or longer**—days 1, 7, 14, 21
2. **Rise** of serum β -hCG level **>10 percent** during **three weekly consecutive measurements** or longer, during a period of **2 weeks or more**—days 1, 7, 14
3. Serum β -hCG level **remains detectable for 6 months or more**
4. Histological criteria for **choriocarcinoma**

Memory hook: "4-3 plateau, 3-2 rise, 6 months still there, or chorio on the slide."

Diagnosis, staging and prognostic scoring

- **Most common finding: irregular bleeding with uterine subinvolution** (continuous, intermittent, or sudden and massive). Myometrial perforation → **intraperitoneal hemorrhage**.
- May show **lower genital tract metastases** (bluish vascular masses) or only distant metastases with no uterine tumor.
- **Thinking of GTN is the most important factor**: unusually **persistent bleeding after any pregnancy** → **measure β -hCG**.
- **⚠ Do not biopsy vaginal metastases** (unnecessary, can bleed heavily).
- **Work-up**: baseline β -hCG, hemogram, **liver and renal function**, **TVS**, **chest CT or radiograph**, **brain and abdominopelvic CT or MRI**; less often **PET** and **CSF β -hCG**.
- **No extrauterine disease**: a **second curettage or hysterectomy** may be considered → β -hCG every **2 weeks** until **3 consecutive undetectable**, then **monthly for 6 months**.
- **Persistent hCG or extrauterine disease** → **stage and start chemotherapy**.
- Staging is **clinical (FIGO 2009)** with the **modified WHO prognostic score** (each factor scored 0–4).
 - **Score 0–6 = low risk; ≥ 7 = high risk.**

Table 13-5. International Federation of Gynecology and Obstetrics (FIGO) Staging and Diagnostic Scoring System for Gestational Trophoblastic Neoplasia**Anatomical Staging**

Stage	Extent
I	Disease confined to the uterus
II	GTN extends outside of the uterus but is limited to the genital structures (adnexa, vagina, broad ligament)
III	GTN extends to the lungs , with or without known genital tract involvement
IV	All other metastatic sites

Modified World Health Organization (WHO) Prognostic Scoring System^a

Scores ^b	0	1	2	4
Age (years)	<40	≥40	—	—
Antecedent pregnancy	Mole	Abortion	Term	—
Interval after index pregnancy (mo)	<4	4–6	7–12	>12
Pretreatment serum β-hCG (mIU/mL)	<10 ³	10 ³ to 10 ⁴	10 ⁴ to 10 ⁵	≥10 ⁵
Largest tumor size (including uterus)	<3 cm	3–4 cm	≥5 cm	—
Site of metastases		Spleen, kidney	GI	Liver, brain
Number of metastases	—	1–4	5–8	>8
Previous failed chemotherapy drugs	—	—	1	≥2

^aAdapted by FIGO. ^bLow risk = WHO score of 0 to 6; high risk = WHO score of ≥7. β-hCG = beta human chorionic gonadotropin; GI = gastrointestinal; GTN = gestational trophoblastic neoplasia.

Histological classification (staging ignores histology)

Type	Origin / features	Behaviour	hCG	Treatment
Invasive mole (chorioadenoma destruens)	Almost all arise from complete or partial moles; trophoblast + whole villi deep in myometrium, sometimes peritoneum, parametrium, vaginal vault	Locally aggressive, less prone to metastasize	Raised	Chemotherapy
Gestational choriocarcinoma	Most common GTN after term pregnancy or miscarriage (only ¼ follow a mole); cyto- and syncytiotrophoblast, no villi ; invades myometrium and vessels → hemorrhage, necrosis, dark surface nodules	Early blood-borne metastases: lungs and vagina most common; also vulva, kidney, liver, brain, ovary, bowel. Theca-lutein cysts common	High	Chemotherapy
Placental site trophoblastic tumor (PSTT)	Uncommon; intermediate trophoblast at the placental site	Locally invasive, chemoresistant	Only modestly raised ; high proportion of free β-hCG favors it	Hysterectomy ; adjuvant multiagent chemo for higher-risk stage I and later stages
Epithelioid trophoblastic tumor (ETT)	Rare; chorionic-type intermediate trophoblast ; mainly uterine	Relatively chemoresistant; metastases common	Low , with bleeding	Hysterectomy ; combination chemo for metastases

Treatment

- Best managed by **oncologists**, ideally in **specialist GTN centers**. **Chemotherapy alone** is usually primary.
- Second evacuation (no extrauterine disease) may avoid or minimize chemo; curettage for bleeding or bulky retained tissue; hysterectomy as primary or adjuvant therapy in specific cases.
- **Low-risk / nonmetastatic: single-agent chemotherapy: methotrexate or actinomycin D** (equally effective as both combined; **methotrexate less toxic**). Repeat until hCG undetectable.

- **High-risk: combination chemotherapy**, e.g. **EMA-CO** (Etoposide, **M**ethotrexate, **A**ctinomycin D, **C**yclophosphamide, **O**ncovin/vincristine) → **cure ~90%**. Adjuvant surgery or radiotherapy in selected cases.
- Causes of death: **hemorrhage from metastases**, respiratory failure, sepsis, multiorgan failure from chemoresistant disease.
- **After hCG is undetectable: surveillance for 1 year** with **effective contraception** (chemo is teratogenic; a new pregnancy confuses hCG).
- **Quiescent hCG**: very low, **plateaued** hCG without metastases (dormant trophoblast) → **observe without treatment**; **20%** later recur with active GTN.

6. Subsequent Pregnancy

- **After a mole**: fertility and outcomes are usually normal. **~1% risk of GTD in a later pregnancy** → **early sonography**.
- **After GTN chemotherapy: delay pregnancy for 1 year** (most relapses occur within it; methotrexate persists in tissues for months).
 - Pregnancy within 1 year usually still has a **favorable outcome**; no ↑ miscarriage or relapse, but relapse **diagnosis may be delayed** by the pregnancy.
 - Fertility and anomaly rates normal; **stillbirth ↑: 1.5% vs 0.8%** background.
- **For any later pregnancy after a mole or GTN**: send the **placenta/products for pathology** and check **β-hCG 6 weeks postpartum** (more useful after treated GTN than after a mole alone).

Quick-Recall: Complete vs Partial Mole vs GTN Types

Feature	Complete mole	Partial mole	Choriocarcinoma	PSTT / ETT
Karyotype	46,XX (all paternal)	69,XXX/XXY (diandric triploid)	Any antecedent pregnancy	Any antecedent pregnancy
Fetus	Absent	Present (abnormal)	—	—
Villi	Yes, diffuse edema	Yes, focal edema	No	No
p57	Negative	Positive	—	—
hCG	>100,000, often millions	Usually as for dates	High	Low / modest (PSTT: free β-hCG)
Uterus	Large for dates	Small for dates	Subinvolution, bleeding	Bleeding
Theca-lutein cysts	25–30%	Rare	Common	—
Progression to GTN	15–20%	1–5%	—	—
Metastases	—	—	Early, blood-borne (lung, vagina)	ETT: common
Treatment	Suction evacuation + hCG follow-up	Suction evacuation + hCG follow-up	Chemotherapy (MTX/actinomycin D; EMA-CO if high risk)	Hysterectomy (chemoresistant)

Key Numbers to Memorize

- Mole incidence **1/1000** deliveries; recurrence **0.9%** (complete), **0.3%** (partial), **~20%** after two complete moles
- Evacuation at median **9 wk** (complete) and **12 wk** (partial)
- Theca-lutein cysts **25–30%** of complete moles; cysts **>6 cm** = GTN risk factor
- Complete mole hCG **>100,000 mIU/mL** (millions possible → hook effect)
- Sonography for mole: sensitivity **44%**, specificity **74%**

- Karman cannula **10–14 mm**; oxytocin **20 U/1000 mL**; methylergonovine **0.2 mg**; carboprost **250 µg**; misoprostol **800–1000 µg** PR
- GTN after complete mole **15–20%**, partial **1–5%**; age **40–49** → **up to 50%**
- hCG baseline within **48 h**, every **1–2 wk** until undetectable, then monthly × **6 months**; resolution median **6–7 wk**; **95%** by 14 wk, **99%** by 25 wk
- GTN origin: **½** mole, **¼** miscarriage/ectopic, **¼** term/preterm
- GTN criteria: plateau **±10%** × **4** values over **3 wk**; rise **>10%** × **3** values over **2 wk**; detectable **≥6 months**; choriocarcinoma histology
- WHO score **0–6 low risk, ≥7 high risk**; FIGO stage III = **lungs**; score 4 for liver/brain metastases, hCG **≥10⁵**, interval **>12 mo**
- High-risk cure with EMA-CO **~90%**; surveillance **1 year** after treatment; quiescent hCG recurs in **20%**
- Later pregnancy: GTD recurrence **1%**; delay **1 year** after chemo; stillbirth **1.5% vs 0.8%**; hCG **6 wk postpartum**